Algebraic Number Theory:Assignment

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A.1 Homework 1

Throughout what follows, let R denote an integral domain with fraction field K.

Exercise A.1.1.

Suppose R is a local domain. Let I be a fractional ideal of R. Prove that I is invertible if and only if it is principal.
Hint: There exist $x \in I$, $y \in I^{-1}$ such that $xy \in R^\times$. Then prove $I = xR$.

Solution: Suppose $x \neq 0$ and $I = xR$ is principal then let $J = x^{-1}R$ and $IJ = R$ implies I is invertible. Conversely, if there exist a fractional ideal J such that $IJ = R$. Then there exist $x_i \in I$ and $y_i \in J$ such that

$$\sum x_i y_i = 1.$$

Because R is local, without loss of generality, we can suppose $x_1 y_1 \in R \backslash \mathfrak{m} = R^\times$. Since $x_1 \in I$, $x_1 R \subseteq I$ is obvious. Hence $y_1 \in J$ implies $y_1 I \subseteq R$, multiplying by x_1 on both sides, we obtain that $I \subseteq x_1 R$. Therefore, $I = x_1 R$ is principal. ■

Exercise A.1.2.

Prove that a Dedekind domain with only finitely many prime ideals is a PID.

Solution: Let $\mathfrak{p}_1, \ldots, \mathfrak{p}_n$ be all prime ideals of Dedekind domain R. It suffices to prove all prime ideals are principal. Without loss of generality, we prove $\mathfrak{p}_1$ is principal and others are in the same way. Let $x \in \mathfrak{p}_1 \backslash \mathfrak{p}_1^2$ and by Chinese Remainder Theorem, there exist $u \in R$ such that

$$u \equiv x \bmod \mathfrak{p}_1^2 \text{ and } u \equiv 1 \bmod \mathfrak{p}_i \text{ for } i \neq 1.$$

Let $v_1, \ldots, v_n$ be the valuations of local rings $R_{\mathfrak{p}_1}, \ldots, R_{\mathfrak{p}_n}$ respectively. Then we obtain that

$$v_1(u) = 1 \text{ and } v_i(u) = 0 \text{ for } i \neq 1,$$

because $u - x \in \mathfrak{p}_1^2$ and implies $u \in (x) + \mathfrak{p}_1^2 \subseteq \mathfrak{p}_1$ meaning that $v_1(u) \geq 1$. Suppose $v_1(u) \geq 2$, equivalently $u \in \mathfrak{p}_1^2$, then $x \in (u) + \mathfrak{p}_1^2 = \mathfrak{p}_1^2$ contradicting our assumption,

therefore $v_1(u) = 1$. Because R is Dedekind domain, the unique factorization of fractional ideals implies

$$(u) = \mathfrak{p}_1^{v_1(u)} \dots \mathfrak{p}_n^{v_n(u)} = \mathfrak{p}_1,$$

the principality of $\mathfrak{p}_1$ follows. ■

Exercise A.1.3.

Suppose R is a DVR with maximal ideal $\mathfrak{p}$. Let $k = R/\mathfrak{p}$ and $U = R^\times$. Let v be the normalized discrete valuation of K associated to R. For each $n \geq 1$, define

$$U_n = 1 + \mathfrak{p}^n = \{x \in K \mid v(x-1) \geq n\}.$$

1. Show that each U_n is a subgroup of U and that $\bigcap_{n=1}^\infty U_n = 1$.
2. Prove that the natural quotient map $R \to k = R/\mathfrak{p}$ induces an isomorphism of multiplicative groups $U/U_1 \cong k^*$.
3. Prove that for each $n \in \mathbb{Z}$, there is an isomorphism of R-modules $k = R/\mathfrak{p} \xrightarrow{\sim} \mathfrak{p}^n/\mathfrak{p}^{n+1}$.
4. Prove that for each $n \geq 1$, the map $u \mapsto u - 1$ induces an isomorphism of abelian groups $U_n/U_{n+1} \xrightarrow{\sim} \mathfrak{p}^n/\mathfrak{p}^{n+1}$.
5. Let p be the characteristic of k. (So $p = 0$ or p is a prime number.)
   (a) Show that $U_n^p := \{a^p \mid a \in U_n\}$ is contained in U_{n+1} for every $n \geq 1$.
   (b) Let $m \geq 1$ be an integer not divisible by p. Prove that for every $n \geq 1$, the group homomorphism
   $$U_n \longrightarrow U_n\ ; \quad u \longmapsto u^m$$
   is injective.

Solution: **(1)** Closed under multiplication holds since $(1+\mathfrak{p}^n)(1+\mathfrak{p}^n) \subseteq 1+\mathfrak{p}^n$. Identity element is 1. Suppose $x \in U_n$, then $v(x^{-1}-1) = v(x^{-1}) + v(1-x)$ hence x is invertible meaning that $v(x) = 0$ and $v(1-x) = v(x-1)$ implies $v(x^{-1}-1) \geq n$. Suppose $x \in \bigcap_{n=1}^\infty U_n$, then we obtain $v(x-1) = \infty$, equivalently, $x = 1$. **(2)** The natural quotient map restricted to U, denoted by $\pi : U \to k^*$, induces isomorphism $U/U_1 \cong k^*$ because $\ker(\pi) = U_1$. **(3)** Consider commutative graph

$$\begin{tikzcd}
0 \arrow[r] & \mathfrak{p} \arrow[r, hook] \arrow[d, "\cdot u^n", "\cong"'] & R \arrow[r, two heads] \arrow[d, "\cdot u^n", "\cong"'] & k \arrow[r] \arrow[d, dashed, "\bar{\phi}", "\cong"'] & 0 \\
0 \arrow[r] & \mathfrak{p}^{n+1} \arrow[r, hook] & \mathfrak{p}^n \arrow[r, two heads] & \mathfrak{p}^n/\mathfrak{p}^{n+1} \arrow[r] & 0
\end{tikzcd}$$

which the third isomorphism follows from the Five Lemma. **(4)** Consider a surjective group homomorphism

$$\varphi : U_n \to \mathfrak{p}^n/\mathfrak{p}^{n+1} \quad u \mapsto u - 1 + \mathfrak{p}^{n+1}$$

with kernel U_{n+1}, then the first isomorphism theorem yields isomorphism $U_n/U_{n+1} \cong \mathfrak{p}^n/\mathfrak{p}^{n+1}$. **(5.a)** For $p = 0$, obvious to see that $U_n^0 = \{1\} \subseteq U_{n+1}$. For $p \neq 0$ and any $a \in U_n$,

$$v(a^p - 1) = v((a-1)^p) = p \cdot v(a-1) \geq n+1$$

implies $a^p \in U_{n+1}$. **(5.b)** It suffices to prove for $m = q$ where q is a prime number and coprime to p. Suppose that $u \in U_n$ such that $u^q = 1$. According to Bézout's identity and **(5.a)**, suppose $ap + bq = 1$, then $u = u^{ap+bq} = u^{ap} \in U_{n+1}$. Thus we can consider a new group homomorphism: $U_{n+1} \to U_{n+1}$. Continuing this process, we obtain that $u \in \bigcap_{n=1}^{\infty} U_n$, then $u = 1$ and injectivity holds. ■

Exercise A.1.4.

Prove that if a domain R is integrally closed, then so is the polynomial ring $R[t]$.

Solution: Let $f = p/q$ be an arbitrary integral element in $R(t)$. Without loss of generality, we can suppose q is irreducible, because if $q = q_1 q_2$ then we consider integral element fq_1, and suppose p, q are coprime. Suppose f satisfies a monic polynomial $f^n + a_{n-1}f^{n-1} + \cdots + a_0 = 0$ where $a_k \in R[t]$. Then, by multiplying both side by q^n , we obtain that $p^n + a_{n-1}qp^{n-1} + \cdots + a_0 q^n = 0$, which implies that $q|p^n$. Let $F = \mathrm{Frac}(R)$ be the fraction field of R, and interprete p, q as elements in $F[t]$, then we obtain that $q|p$ since $F[t]$ is an UFD and hence $q \in F^*$ because p, q coprime. Therefore, we can write $f = c_m t^m + \cdots + c_0$ where $c_k \in F$. Let $f_1 = t^r - f$, where $r = \deg f + \sum \deg a_k$, then f_1 is the root of polynomial

$$P(X) = (t^r - X)^n + a_{n-1}(t^r - X)^{n-1} + \cdots + a_0.$$

Let $X = 0$, we obtain the constant term of this polynomial

$$t^{rn} + a_{n-1}t^{rn-r} + a_{n-2}t^{rn-2r} + \cdots + a_0$$

denoted by $P(0)$. Let $P(X) = XQ(X) + P(0)$, then from $P(f_1) = 0$ we obtain that

$$P(0) = -f_1 Q(f_1),$$

the roots of f_1 are among the roots of monic polynomial $P(0) \in R[t]$, the coefficients of f as of f_1 are elementary symmetric polynomials of its roots, which are integral over R. Therefore, we obtain that the coefficients of f are contained in R, $R[t]$ is integrally closed. ■

Exercise A.1.5.

Let $A = \mathbb{Q}[X, Y]/(X^2 - Y^3)$. Show that A is a domain but it is not integrally closed.

Solution: $(X^2 - Y^3)$ is a prime ideal because $X^2 - Y^3$ is irreducible and $\mathbb{Q}[X, Y]$ is UFD, *a fortiori*, A is a domain. Hence, X/Y is the root of a monic polynomial $P(Z) = Z^2 - Y$ but not contained in A. If it is containd, then there exist $f, g \in \mathbb{Q}[X, Y]$ such that

$$X = Yf - (X^2 - Y^3)g$$

implies contradiction by comparing their linear terms. ■

Exercise A.1.6.

Let R be a Dedekind domain and let $\mathfrak{a} \subseteq R$ be a nonzero ideal. Show that the quotient ring $R/\mathfrak{a}$ has only finitely many ideals and that all of these ideals are principal.

Solution: By isomorphism theorem of rings, the natural quotient $R \to R/\mathfrak{a}$ gives one-to-one correspondence of the ideals of R containing $\mathfrak{a}$ and the ideals of $R/\mathfrak{a}$. Therefore, all ideals of $R/\mathfrak{a}$ can be written as the form

$$\mathfrak{p}_1^{i_1} \dots \mathfrak{p}_n^{i_n} + \mathfrak{a}$$

where $\mathfrak{p}_k$ are the prime ideals of R containing $\mathfrak{a}$ and $0 \leq i_k \leq v_k(\mathfrak{a})$ where v_k is the valuation associated to $\mathfrak{p}_k$. The principality directly from *Exercise A.1.2.* ■

Exercise A.1.7.

Show that every integral ideal of a Dedekind domain can be generated by two elements.

Solution: Let $\mathfrak{a}$ is an arbitrary ideal of R with prime ideal decomposition

$$\mathfrak{a} = \mathfrak{p}^k \mathfrak{b}$$

Where $\mathfrak{p}$ is a prime ideal coprime to the ideal $\mathfrak{b}$. The quotient ring $R/\mathfrak{p}^{k+1}\mathfrak{b}$ is principal from *Exercise A.1.6.* Therefore we can assume that $\mathfrak{a} = (a) + \mathfrak{p}^{k+1}\mathfrak{b}$. Hence, consider another quotient ring $R/(a)$, we obtain that $\mathfrak{a} = (a, b)$. ■

A.2 Homework 2

Exercise A.2.1

Let R be the ring of algebraic integers in $\mathbb{C}$, i.e., the integral closure of $\mathbb{Z}$ in $\mathbb{C}$. Consider the ideal $I \subseteq R$ generated by the infinite subset $\{ \sqrt[2^n]{2} \mid n \in \mathbb{N}\}$.

Show that I is not finitely generated. Deduce that R is not Noetherian (and hence not a Dedekind domain).

Solution: Suppose that I is finitely generated with generators $\alpha_1, \dots, \alpha_m$. According to Chevalley's Extension Theorem for Valuations, we can extend 2-adic valuation on $\mathbb{Q}$ to $\overline{\mathbb{Q}}$, let v_2 denote this extended valuation. By definition, we obtain that $R \subseteq \overline{\mathbb{Q}}$ and $v_2(R) \geq 0$. Since I is generated by $\{ \sqrt[2^n]{2}\}$ and hence $v_2(\sqrt[2^n]{2}) = 2^{-n}$ is strictly greater than zero, the valuation $v_2(\alpha_i)$ is strictly greater than zero. Let $\varepsilon = \min\{v_2(\alpha_i)\}$ be the lower bound of valuations of finite generators α_i. For arbitrary $r \in I$, we can write $r = \sum_{i=1}^m c_i\alpha_i$ for some $c_i \in R$. Using the non-Archimedean property and $v_2(c_i) \geq 0$, we obtain:

$$v_2(r) \geq \min_i\{v_2(c_i) + v_2(\alpha_i)\} \geq \min_i\{v_2(\alpha_i)\} = \varepsilon > 0$$

However we find that $v_2(\sqrt[2^n]{2}) = 2^{-n}$ converging to zero, which is a contradiction. ■

Exercise A.2.2

Let R denote a Dedekind domain with fraction field K and suppose $R \neq K$.

Prove that there exists a set Ω of normalized discrete valuations of K such that the following two properties hold:

(R1) $R = \{x \in K \mid v(x) \geq 0 \text{ for all } v \in \Omega\}$;

(R2) for every $a \in K^\times$, the set $\{v \in \Omega \mid v(a) \neq 0\}$ is finite.

Solution: Let

$$\Omega = \{v_\mathfrak{p} \mid \mathfrak{p} \in \mathrm{Spm}(R)\}$$

where $v_\mathfrak{p}$ is the normalized discrete valuation associated with $R_\mathfrak{p}$. For (R1), we find that $v_\mathfrak{p}(x) \geq 0$ if and only if $x \in R_\mathfrak{p}$, then we obtain that

$$\{x \in K \mid v(x) \geq 0 \text{ for all } v \in \Omega\} = \bigcap_{\mathfrak{p} \in \mathrm{Spm}(R)} R_\mathfrak{p} = R.$$

For $(R2)$, we first fix an arbitrary element $a \in K^*$ and let A denote the set $\{v \in \Omega \mid v(a) \neq 0\}$. Since R is Dedekind, we may suppose that

$$aR = \mathfrak{p}_1^{a_1} \dots \mathfrak{p}_n^{a_n}.$$

where $a_i \neq 0$. I claim that $A = \{v_{\mathfrak{p}_1}, \dots, v_{\mathfrak{p}_n}\}$. For an arbitrary maximal ideal $\mathfrak{q} \notin \{\mathfrak{p}_1, \dots, \mathfrak{p}_n\}$, it does not appear in the factorization of aR, yielding $v_\mathfrak{q}(a) = 0$ and hence $v_\mathfrak{q} \notin A$. This implies $A \subseteq \{v_{\mathfrak{p}_1}, \dots, v_{\mathfrak{p}_n}\}$. Conversely, by definition we have $v_{\mathfrak{p}_i}(a) = a_i \neq 0$, meaning $\{v_{\mathfrak{p}_1}, \dots, v_{\mathfrak{p}_n}\} \subseteq A$. Therefore, A is precisely this finite set. ■

Exercise A.2.3

Let R denote an integral domain with fraction field K. Let Ω be a set of normalized discrete valuations of K satisfying Properties (R1) and (R2) in Exercise A.2.2. For each $v \in \Omega$, let R_v be its valuation ring and let $\mathfrak{p}_v$ be the maximal ideal of R_v.

Suppose further that every nonzero prime ideal of R is maximal.

1. Let $\mathfrak{p}$ be a maximal ideal of R.
   (a) Let $z \in K^\times$ be such that $R \cap z^{-1}R \subseteq \mathfrak{p}$. Show that there exists $v \in \Omega$ such that $\mathfrak{p} = R \cap \mathfrak{p}_v$ and $v(z) < 0$.
   (b) Prove that $\Omega_\mathfrak{p} := \{v \in \Omega \mid R_v \supseteq R_\mathfrak{p}\}$ is a finite set.
   (c) Prove that $R_\mathfrak{p} = R_v$ for a unique $v \in \Omega$. Thus $\Omega_\mathfrak{p}$ consists of a single element.
2. Prove that for every $v \in \Omega$, there exists a unique maximal ideal $\mathfrak{p}$ such that $R_v = R_\mathfrak{p}$.
3. Deduce from (1) and (2) that the map

$$\Omega \longrightarrow \mathrm{Spm}(R) \ ; \ v \longmapsto \mathfrak{p}_v \cap R$$

   is well defined and bijective. Let $\mathfrak{p} \mapsto v_\mathfrak{p}$ denote its inverse map.

4. Let $I \subseteq R$ be a nonzero ideal.
   (a) Prove that $\{\mathfrak{p} \in \mathrm{Spm}(R) \mid v_{\mathfrak{p}}(I) \neq 0\}$ is finite, where $v_{\mathfrak{p}}(I) := \min\{v_{\mathfrak{p}}(x) \mid x \in I\}$.
   (b) Prove that there is a unique factorization $I = \prod_{\mathfrak{p} \in \mathrm{Spm}(R)} \mathfrak{p}^{r_{\mathfrak{p}}}$, where each $r_{\mathfrak{p}} \in \mathbb{N}$ and almost all of them are zero.
   (c) If $J \subseteq R$ is another nonzero ideal, show that $I \mid J$ if and only if $I \supseteq J$.
5. Prove that R is a Dedekind domain.

Solution: (1.a) First I claim that there exists $v \in \Omega$ such that $v(z) < 0$. If not, then for every $v \in \Omega$ we have $v(z) \geq 0$. In this case, we have $v(zR_z) \geq 0$ and therefore

$$R_v \subseteq z^{-1}R_v.$$

Hence we find that

$$\mathfrak{p} \supseteq R \cap z^{-1}R = R \cap \left(\bigcap_{v \in \Omega} z^{-1}R_v \right) \supseteq R \cap \left(\bigcap_{v \in \Omega} R_v \right) = R$$

which is a contradiction. Let

$$A = \{v \in \Omega \mid v(z) < 0\}.$$

By (R2), A is a finite set, then we say $A = \{v_1, \ldots, v_n\}$. We suppose that $v_i(z) = -a_i$, then we find that

$$R \cap z^{-1}R = \{x \in K \mid v_i(x) \geq a_i, v(x) \geq 0 \text{ for any } v \in \Omega\}.$$

And therefore

$$R \cap z^{-1}R = \left(\bigcap_{v_i \in A} R \cap \mathfrak{p}_{v_i}^{a_i} \right).$$

Since the product of ideals is always contained in their intersection, we have:

$$\prod_{v_i \in A} (R \cap \mathfrak{p}_{v_i})^{a_i} \subseteq \bigcap_{v_i \in A} (R \cap \mathfrak{p}_{v_i}^{a_i}) = R \cap z^{-1}R \subseteq \mathfrak{p}$$

Thus, there exists some i such that $R \cap \mathfrak{p}_{v_i} \subseteq \mathfrak{p}$. Because v_i is non-trivial, $R \cap \mathfrak{p}_{v_i}$ is a non-zero prime ideal, and hence maximal by our assumption. Since $\mathfrak{p}$ is also maximal, we must have exactly:

$$\mathfrak{p} = R \cap \mathfrak{p}_{v_i}$$

(1.b) Suppose that $\Omega_{\mathfrak{p}}$ is an infinite set and for arbitrary $v \in \Omega_{\mathfrak{p}}$ we have

$$\mathfrak{p}_v \cap R_{\mathfrak{p}} = \mathfrak{p}R_{\mathfrak{p}}.$$

Since $\mathfrak{p}_v \cap R$ is a prime ideal and therefore it is maximal. $\mathfrak{p}_v \cap R_{\mathfrak{p}}$ is the prime ideal of $R_{\mathfrak{p}}$ lying over $\mathfrak{p}_v \cap R$, then we have $\mathfrak{p}_v \cap R_{\mathfrak{p}}$ is also maximal implying what we want. Therefore

$$\bigcap_{v \in \Omega_{\mathfrak{p}}} \mathfrak{p}_v \supseteq \mathfrak{p}R_{\mathfrak{p}}$$

For arbitrary nonzero element $x \in \mathfrak{p}R_\mathfrak{p}$ and $v \in \Omega_\mathfrak{p}$, we have $v(x) \neq 0$ which contradicts to (R2) since we suppose $\Omega_\mathfrak{p}$ is an infinite set.

(1.c) First I claim that

$$R_\mathfrak{p} = \bigcap_{v \in \Omega_\mathfrak{p}} R_v.$$

It is easy to see that $R_\mathfrak{p} \subseteq \bigcap_{v \in \Omega_\mathfrak{p}} R_v$ by definition. Conversely, suppose that $x \in \bigcap_{v \in \Omega_\mathfrak{p}} R_v$. Then we obtain that $v'(x) \geq 0$ for arbitrary $v' \in \Omega_\mathfrak{p}$. For arbitrary $v \in (\Omega \setminus \Omega_\mathfrak{p})$, we have $\mathfrak{p} \neq R \cap \mathfrak{p}_v$. If not, v must contained in $\Omega_\mathfrak{p}$. Therefore, by (1.a), we find that $R \cap x^{-1}R$ is not containd in $\mathfrak{p}$. Suppose that $y \in (R \cap x^{-1}R) - \mathfrak{p}$, then we find for arbitrary $v \in \Omega$, $v(y), v(xy) \geq 0$ and for arbitrary $v' \in \Omega_\mathfrak{p}$, we have $v'(y) = 0$. It means that $xy \in R \subseteq R_\mathfrak{p}$ and $y \in R - \mathfrak{p}$. Since $v'(R - \mathfrak{p}) = v'((R_{v'} - \mathfrak{p}_{v'}) \cap R) = 0$ and $v'(\mathfrak{p}) = v'(\mathfrak{p}_{v'} \cap R) \geq 1$, then it must have $y \in R - \mathfrak{p}$, therefore $x \in R_\mathfrak{p}$ meaning that $R_\mathfrak{p} \supseteq \bigcap_{v \in \Omega_\mathfrak{p}} R_v$.

Select an arbitrary valuation $v \in \Omega_\mathfrak{p}$, I claim that $R_v = R_\mathfrak{p}$. From the discussion of (1.b), we obtain that

$$v_\mathfrak{p} \cap R_\mathfrak{p} = \mathfrak{p}R_p.$$

It means that $v(R_\mathfrak{p} - \mathfrak{p}R_p) = 0$. Hence, we have

$$v_\mathfrak{p}^2 \cap R_\mathfrak{p} = \mathfrak{p}^2 R_p.$$

We notice that $v_\mathfrak{p}^2 \cap R_\mathfrak{p} \supseteq \mathfrak{p}^2 R_p$ since $v(\mathfrak{p}^2 R_\mathfrak{p}) \geq v(\mathfrak{p}^2) \geq 2$. Therefore, $v_\mathfrak{p}^2 \cap R_\mathfrak{p}$ must equals $\mathfrak{p}^2 R_\mathfrak{p}$ or $\mathfrak{p}R_\mathfrak{p}$. If it equals $\mathfrak{p}R_\mathfrak{p}$, then for arbitrary $x \in K^*$, if $v(x) = 1$ then we have $x \notin R_\mathfrak{p}$, which is wrong. To see that, we choose an arbitrary element $x \in K^*$ such that $v(x) = 1$. Suppose that $\Omega_\mathfrak{p} = \{v, v_1, \ldots, v_k\}$. Then we use the weak approximation theorem, there exists an element $y \in K^*$ such that $v(y) = 0$ and $v_i(y) = -v_i(x)$. Then we find that $v(xy) = 1$ and $v_i(xy) = 0$ meaning that $xy \in R_\mathfrak{p}$. Therefore, it must have $v_\mathfrak{p}^2 \cap R_\mathfrak{p} = \mathfrak{p}^2 R_\mathfrak{p}$. Use same method, we can prove that $v_\mathfrak{p}^j \cap R_\mathfrak{p} = \mathfrak{p}^j R_\mathfrak{p}$ for any positive integer j. It meaning that $R_v = R_\mathfrak{p}$. Since select arbitrarily, the uniqueness is obtained.

(2) Let $\mathfrak{p} = \mathfrak{p}_v \cap R$. It is a prime ideal and hence is maximal ideal. We find that $v(R - \mathfrak{p}) = 0$ and therefore $v(R_\mathfrak{p}) \geq 0$ implying $R_p \subseteq R_v$. By (1), we have $R_\mathfrak{p} = R_v$. If there exists a different maximal ideal $\mathfrak{q}$ such that $R_v = R_\mathfrak{q}$. Then it means that $\mathfrak{p}_v$ is lying over $\mathfrak{q}$, then it must equal to $\mathfrak{p}$.

(3) We just need to prove that $R_\mathfrak{p} \subseteq R_v$ according to (1). It is quite clear from the construction of (2).

(4.a) We notice that $v_\mathfrak{p}(I) \neq 0$ if and only if $I \subseteq \mathfrak{p}$. If set $\{\mathfrak{p} \in \mathrm{Spm}(R) \mid v_\mathfrak{p}(I) \neq 0\}$ is not finite, It means that for arbitrary element $x \in I$, there exists infinite valuations v such that $v(x) \neq 0$. Then by (R2) we must have $I = (0)$ contradicting to our condition.

(4.b) Let J denote $\prod_{i=1}^n \mathfrak{p}^{r_{\mathfrak{p}_i}}$ such that for $\mathfrak{p} \notin \{\mathfrak{p}_1, \ldots, \mathfrak{p}_n\}$ we have $v_\mathfrak{p}(I) = 0$ and $r_\mathfrak{p} = v_\mathfrak{p}(I)$. We notice that $I_\mathfrak{p} = \mathfrak{p}^{r_\mathfrak{p}} R_\mathfrak{p}$ since

$$I_\mathfrak{p} \subseteq \mathfrak{p}^{r_\mathfrak{p}} R_\mathfrak{p} \text{ and } I_\mathfrak{p} \subsetneq \mathfrak{p}^{r_\mathfrak{p}+1} R_\mathfrak{p}$$

Then we have

$$I = \bigcap_{\mathfrak{p} \in \mathrm{Spm}(R)} I_\mathfrak{p} = \bigcap_{i=1}^n I_{\mathfrak{p}_i} = J.$$

The last step is because $\mathfrak{p}_i$ are pairwise coprime. (4.c) It is directly deduced by (4.b).

(5) Noetherian is due to (4.c). For intergrality, we consider ideal (x) generated by an arbitrary element $x \in R$. ■